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Optimizing Thermal Maturity Assessment in Unconventional Reservoirs: A Comparative Analysis of Different Maturity Estimation Methods

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Abstract

This study explores different methods of assessing the thermal maturity of source rocks in unconventional reservoirs, with a primary focus on establishing a robust workflow for accurately estimating thermal maturity in highly mature samples. Twenty-Nine source rock and seven flowback fluid samples were collected from 4 wells targeting a Lower Silurian-age Formation. All samples were analyzed using several analytical approaches including: geochemical, optical and fluid compositional analyses. Data generated include graptolite reflectance measurements, pyrolysis Tmax, aromatic hydrocarbon concentrations, and Pressure-Volume-Temperature (PVT) compositional data. To assess the applicability and limitations of each method in evaluating thermal maturity in samples at high maturity range (%VRE > 1.2%), multiple published equations were applied to each dataset to derive thermal maturity values.

Based on the calculated thermal maturity results, significant variation in values was observed depending on the adopted analytical approach. T-max based maturity estimates underestimate maturity levels in rock samples relative to other tested methods, leading to inaccurate representation of actual maturity levels in this area. This observation aligns with previous studies highlighting the suppression of Tmax due to the influence of fluids retention in this formation. Conversely, graptolite reflectance measurement conducted on rock samples revealed higher maturity values, with a difference of up to 0.3% in comparison to Tmax-derived estimates. For fluid samples, maturity values based on the slope factor of $C_3 - nC_5$ volatile light hydrocarbons agreed with fluid typing based on gas-to-oil ratio, thus they were utilized as a baseline to assess the variability from other methods. Interestingly, aromatic-based ratios demonstrated remarkable versatility, producing similar maturity values derived from both optical and fluid compositional data analyses, confirming the reliability and consistency of this method in multiple sample types. This study highlights both the utility and limitations of different methods for assessing thermal maturity in source rock extracts and generated fluids. Among the tested methods, aromatic hydrocarbons emerged as a reliable and versatile indicator of thermal maturity across all tested samples. Utilizing multiple methods can ultimately lead to accurate assessment of thermal maturity in shale source rocks, which is crucial in ensuring the success of unconventional exploration strategies and optimized production planning.

Introduction

Accurate assessment of thermal maturity in source rocks plays a critical role in understanding the behavior and distribution of hydrocarbons within unconventional reservoirs. Various methods have been developed to estimate thermal maturity, each with its own strengths and limitations. The accuracy and reliability of these methods can vary depending on many factors such as depositional environment, thermal maturity, mineral matrix, mud additives and many more. Conflicting reported values from each method can lead to significant uncertainties regarding the exact maturity window and the associated fluid types, ultimately affecting the accuracy of hydrocarbons resources assessments.

While significant progress has been made in understanding thermal maturity trends in the Lower Silurian-age Formation, there remains a knowledge gap in the lateral variability of hydrocarbon thermal maturity and distribution across different parts of the Eastern Arabian basin. By analyzing rock and fluid samples, this study aims to establish a consistent and reliable workflow for estimating thermal maturity in highly mature source rock samples, and derive insights into the geological processes that have influenced hydrocarbon generation, migration, and accumulation. A collection of whole rock samples, rock extracts and produced fluids from several wells targeting the Lower Silurian-age Formation were geochemically analyzed. Initial findings of these analyses are discussed in this paper using an integrated approach with a focus on biomarkers.

Methodology

Sample Inventory

This study utilized 29 core samples retrieved from three wells A, B, and C, all drilled using Water-Based Mud (WBM). The approximate locations of these wells are shown in [Figure 1](#). Each sample was sub-sampled three times to conduct source rock screening, optical measurements and biomarker analysis. Additionally, seven produced fluid samples were recovered from well A_1 during flowback at various intervals. These fluid samples were analyzed to compare the results from rock and fluid samples in well A. [Table 1](#) provides an inventory of the samples and details samples type, depth and quantity for each well.

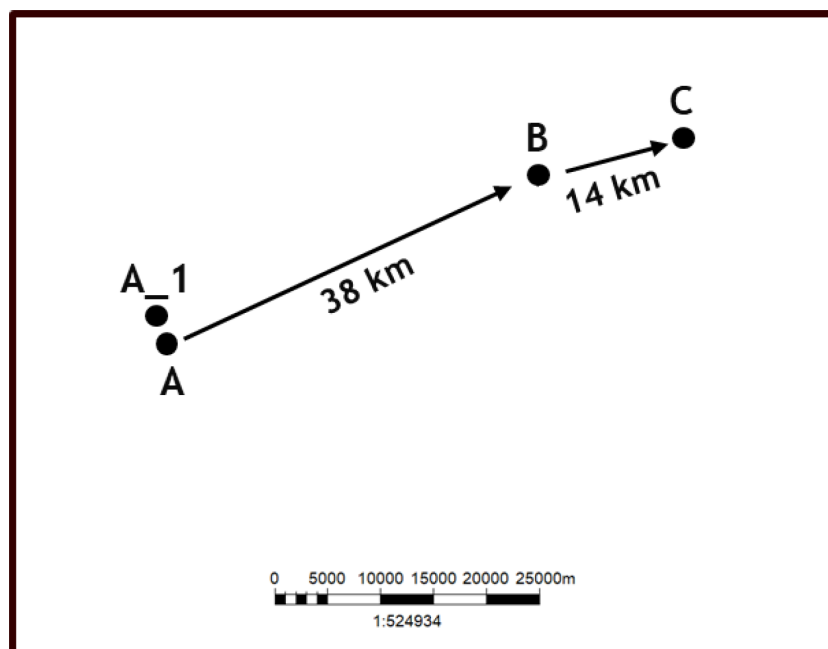


Figure 1—Map showing the approximate locations of wells utilized in this study

Table 1—Shows well names, sample type, normalized sample depth, and the number of tested samples.

<i>Well</i>	<i>Sample Type</i>	<i>Normalized sample depth</i>	<i>N.of samples</i>
A	Rock	220 – 400 ft	13
A_1	Fluid	-	7
B	Rock	1100 – 1200 ft	10
C	Rock	1550 – 1600 ft	6

Source Rock Screening

To conduct the pyrolytic screening, whole rock samples were pulverized using a mortar and a pestle into a 0.212 mm mesh, then were analyzed using a commercial source rock screening instrument. Helium was used as the carrier gas. The initial temperature was set at 300 °C for 3 min, then ramped up to 650 °C at 25 °C /min. The pyrolytic parameters TOC, S1, S2, S3 and Tmax values were measured throughout the programmed pyrolysis. Subsequently, vitrinite reflectance equivalent (%VRE) values for the Lower Silurian-age Formation were calculated using Tmax by applying the following empirical equation from Inan et al. (2016):

$$\%VRE(T_{max}) = 0.0134(T_{max}) - 4.9706 \quad (1)$$

Biomarker Analysis

Core samples were pulverized and then extracted using Accelerated Solvent Extraction (ASE) with Dichloromethane (DCM) as the solvent. Rock extracts and fluid samples were then separated into two fractions, saturate and aromatic, using silica-gel column chromatography. Saturated hydrocarbons were eluted with hexane, while Dichloromethane (DCM) was used to elute the aromatic fraction. The obtained fractions were then concentrated to around 1 mL using a nitrogen blowdown instrument to remove the excess solvent.

For a detailed biomarker analysis, the saturate and aromatic fractions were examined using gas chromatography equipped with Triple Quadrupole Mass Spectrometry on an Agilent 7890B GC/ 7000D MS system. The analysis was conducted with a DB-5 MS fused silica capillary column (60 m × 0.25 mm i.d., film thickness: 0.25 µm). Helium was used as the carrier gas with a flow rate of 1.3 mL/min. For the aromatic fractions, the initial temperature was set at 45 °C for 5 min, ramped up to 310 °C at 4 °C/min, and then maintained at 310 °C for 10 min. The targeted aromatic compounds were detected using the following ions: Methylanthalene (m/z 142), Ethylnaphthalene (m/z 156), Dimethylnaphthalenes (m/z 156), Trimethylnaphthalenes (m/z 156), Phenanthrene (m/z 178), Dibenzothiophene (m/z 184), Methylphenanthrene (m/z 192), Methylthiophene (m/z 198), Dimethylphenanthrene (m/z 206), Ethylphenanthrene (m/z 206), and C2-Dibenzothiophene (m/z 212). The obtained peak areas were used to generate several ratios to investigate the thermal maturity of the tested samples.

Optical Measurements (Graptolite)

Due to the absence of high-land plants remnants and therefore vitrinite macerals in Silurian formations, graptolite reflectance (%GRo) was utilized as a proxy expressed as vitrinite reflectance equivalent (%VRE). Polished thin sections were prepared from core samples. Graptolite fragments were identified under reflected light microscopy using oil immersion at 50x magnification. Measurements were conducted on well-preserved fragments that were encountered at orientations parallel and intermediate to the bedding plane to ensure reliable GRo measurements. For each sample, a minimum of 20 – 50 measurements were taken to ensure statistical representation. The mean value of these measurements (%GRo) was then acquired. To estimate the thermal maturity of the studied rock samples, %GRo values were then converted to %VRE using the correlation of Petersen et al. (2013).

$$\%VRE(GRo) = 0.73(\%GRo) + 0.16 \quad (2)$$

Fluid compositional Data

Pressure-volume-temperature (PVT) compositional data from fluid samples was analyzed through Gas Chromatography (GC) using an in-house Agilent GC system. The molecular composition of hydrocarbons up to C36 was obtained. However, for the purposes of this paper only C1-C5 were utilized. Samples properties such as: API gravity, gas-oil ratio (GOR) and condensate-gas ratio (CGR) were obtained from the recombined well stream samples. These values were used to obtain fluid type of the produced fluids using classifications published by [McCain \(1990\)](#). To assess the thermal maturity of the tested samples, three thermal maturity proxies were utilized: the slope factor (SF) for the molar concentrations of C3-nC5 and the isomer ratios of C4 and C5. To convert the obtained values to vitrinite reflectance equivalent (%VRE), the following empirical equations developed by [Arouri \(2024\)](#) were utilized:

$$\%VRE(SF) = 0.38SF + 0.41 \quad (3)$$

$$\%VRE(i4) = 1.70 \left(\frac{iC_4}{nC_4} \right) + 0.61 \quad (4)$$

$$\%VRE(i5) = 0.89 \left(\frac{iC_5}{nC_5} \right) + 0.56 \quad (5)$$

Results and Discussion

Source Rock Screening

Source rock screening results indicate varying levels of maturity across the analyzed samples, with Tmax values ranging from 438 °C to 485 °C. The majority of the tested samples exhibited Tmax values above 455 °C, indicating late to overmature window. On average, well A samples are 458°C, Well B is 462 °C, while well C samples are 481 °C. Maturity trends between these wells are supported by the decreasing Hydrogen Index (HI) values from well A to C, going from an average of 100 mgHC/gTOC in well A to 23 mgHC/gTOC in well C.

Although pyrolysis results can serve as a good indicator of the general maturity window of the tested samples, some inconsistencies arise once the values are converted to %VRE. For well A, the 10 tested samples ranged in value from 1.10 to 1.27 %VRE placing this well in the late oil window. Well B demonstrated a wider range of values going from 1.13 to 1.37 %VRE, indicating a transition from the late oil to wet gas window. Well C samples, on the other hand, displayed the highest %VRE values ranging from 1.39 to 1.53 %VRE, placing the well within the late wet gas window. It is worth noting that Tmax-based estimate can be affected by multiple factors including mineral matrix, Tmax suppression and fluid retention as highlighted by [Shengyu and Horsfield \(2020\)](#). This can lead in some cases to an underestimation of thermal maturity, and generate a wide range of values within the same well, leading to uncertainties regarding the exact maturity window in a certain location.

Thermal maturity based on Graptolite Reflectance

Thermal maturity values based on Graptolite reflectance showed a wide range of values from 1.11 to 1.76 %VRE across all tested samples. Graptolite-based maturity values showed moderate correlation with Tmax based estimates, with an R^2 of 0.49, as shown in [Figure 2](#). Thermal maturity trends based on Graptolite measurements are consistent with the results, suggesting that both methods are capturing the regional trend accurately. However, most tested samples demonstrated higher maturity values than those based on pyrolysis results, suggesting that Tmax might be underestimating maturity values. This is reflected in the maturity window of the tested wells. Well A samples indicated late oil window maturity, well B is within the wet gas window, while well C is late wet gas to dry gas window. The prediction of fluid types based on Tmax

can be misleading in this case, leading to uncertainties regarding fluid prediction and resources estimation. Flowback results, basin modelling and biomarker analysis in this area suggest that Graptolite-based maturity estimate are more reliable maturity indicators when compared to Tmax-based estimate.

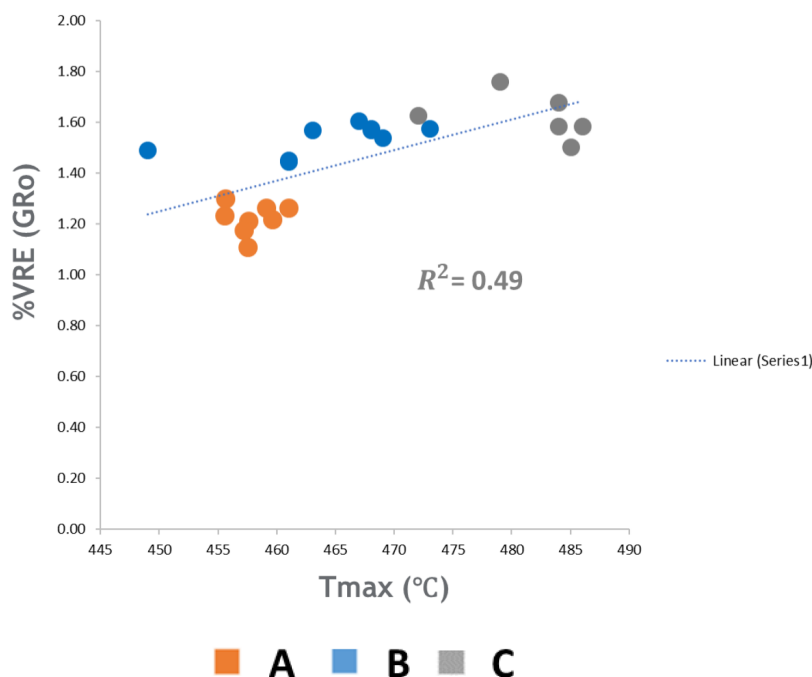


Figure 2—Correlation between %VRE (GRo) and pyrolysis Tmax

PVT maturity proxies

Based on the classification published by McCain (1990), the produced fluids from well A_1 were determined to be within the gas-condensate window. Fluid typing based on GOR will serve as a baseline to validate maturity values obtained from the slope factor, iC_4/nC_4 and iC_5/nC_5 . The slope factor of the tested wellstream sample was calculated based on the exponential equation published by Thompson (2016):

$$y = A(e^{-ax}) \quad (6)$$

The slope factor of $C_3 - nC_5$ molar concentrations was obtained from the compositional profile of well A_1, with a value of 2.78 and an R^2 of 0.999. The calculated %VRE values was 1.47% matching the fluid type based on GOR classification. In contrast, %VRE calculation based on iC_4/nC_4 and iC_5/nC_5 , demonstrated lower maturity values with 1.28 and 1.37 %VRE, respectively. The discrepancy between the obtained values is beyond the scope of this paper. However, based on prior knowledge of the tested area, the SF-based maturity was deemed more reliable.

Aromatic Hydrocarbons as Maturity Indicator

To conduct biomarker analysis, both saturated and aromatic fractions were tested using GC-MS, allowing for the assessment of peak areas for various compounds. Biomarkers derived from saturated and aromatic hydrocarbons have varying applications depending on their thermal stability. A key distinction between these hydrocarbon classes lies in their thermal degradation behavior. Specifically, maturity parameters based on saturated hydrocarbons become less reliable at higher levels of thermal maturity (%VRE > 1.3-1.4%) due to their lower thermal stability (Chen et al., 1996). In the tested samples, saturated biomarkers showed inconsistent results, underscoring their limited utility as maturity indicators at high maturity levels. In contrast, biomarkers based on aromatic hydrocarbons provided more reliable results, confirming their robustness as maturity indicators.

In this study, several aromatic-based ratios were used to calculate maturity values, including the Methyl-naphthalene Ratio (MNR), Trimethyl-naphthalene Ratio (TNR-2), Methylphenanthrene Ratio (MPR), Methylphenanthrene Index-1 (MPI-1), Methylphenanthrene Index-2 (MPI-2), and Methyl-dibenzothiophene Ratio (MDR) (Radke 1988 and references therein; Van Aarssen et al. 1999; Srinivasan et al., 2023). MDR was excluded from maturity analysis in fluids samples, due to the absence of clear Methyl-dibenzothiophene peaks in the tested samples.

The selected ratios were systematically ranked based on three criteria: data dispersion, agreement with other datasets, and relevance for oil samples. Box plots were generated for each ratio to evaluate the statistical distribution of results across samples from the same well. As illustrated in Figure 3, the Naphthalene-based ratios (MNR, TNR-2) demonstrated good overall distribution but still showed significant variability in at least one well. The phenanthrene-based ratios (MPR, MPI-1) exhibited consistent distributions across all tested wells. The dibenzothiophene-based ratio (MDR) showed the highest degree of statistical variability, with a wide spread of values across the wells.

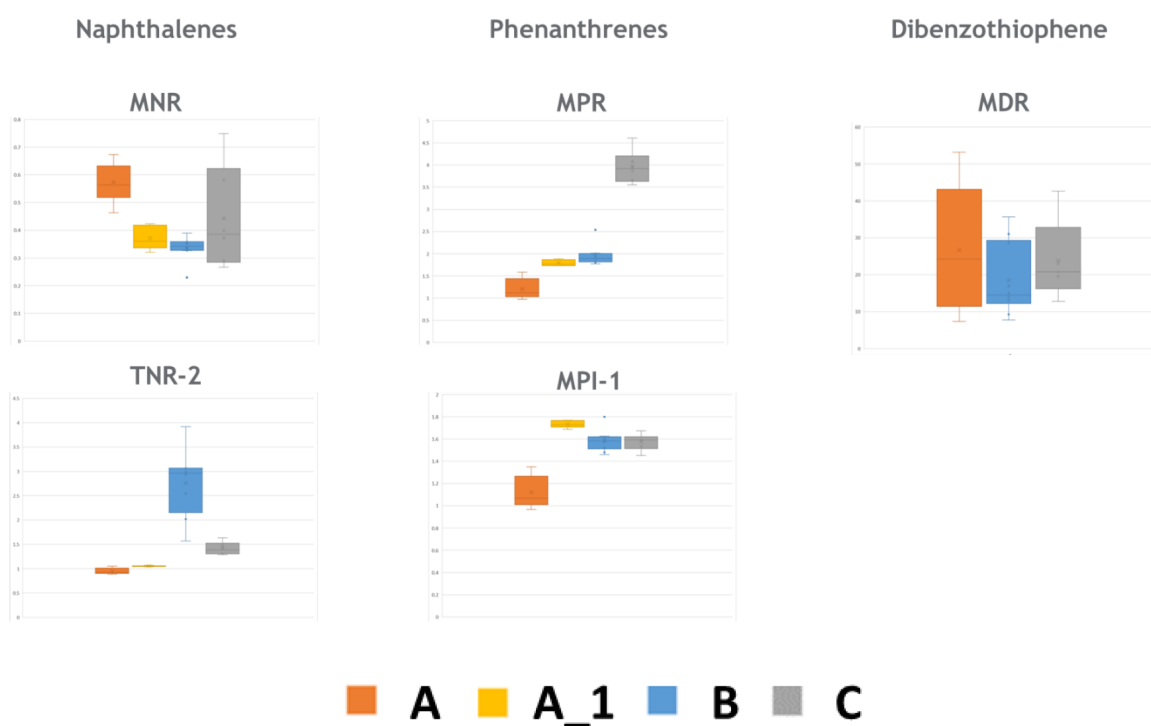


Figure 3—Box plots of selected aromatic ratios showing the statistical spread of results for samples from the same well.

The next step involved ranking these ratios based on their correlation with previously obtained data, specifically depth and pyrolysis Tmax, to evaluate their effectiveness as maturity indicators. Depth was chosen as a correlation parameter since the tested wells showed similar modeled burial history and geothermal gradient. Fluid samples results were excluded from this part of the analysis due to the difficulty in determining the exact depth of the produced fluid. In this analysis, the phenanthrene-based ratios (MPR, MPI-1, and MPI-2) demonstrated the strongest correlations with depth, with R^2 values ranging from 0.74 to 0.78. In contrast, the naphthalene-based ratios (MNR and TNR-2) showed moderate correlations with depth, with R^2 values of 0.36 for MNR and 0.3 for TNR-2, which were influenced by the previously highlighted variability in the obtained values. The dibenzothiophene-based ratio (MDR) showed no clear relationship with depth, as shown in Figure 4.

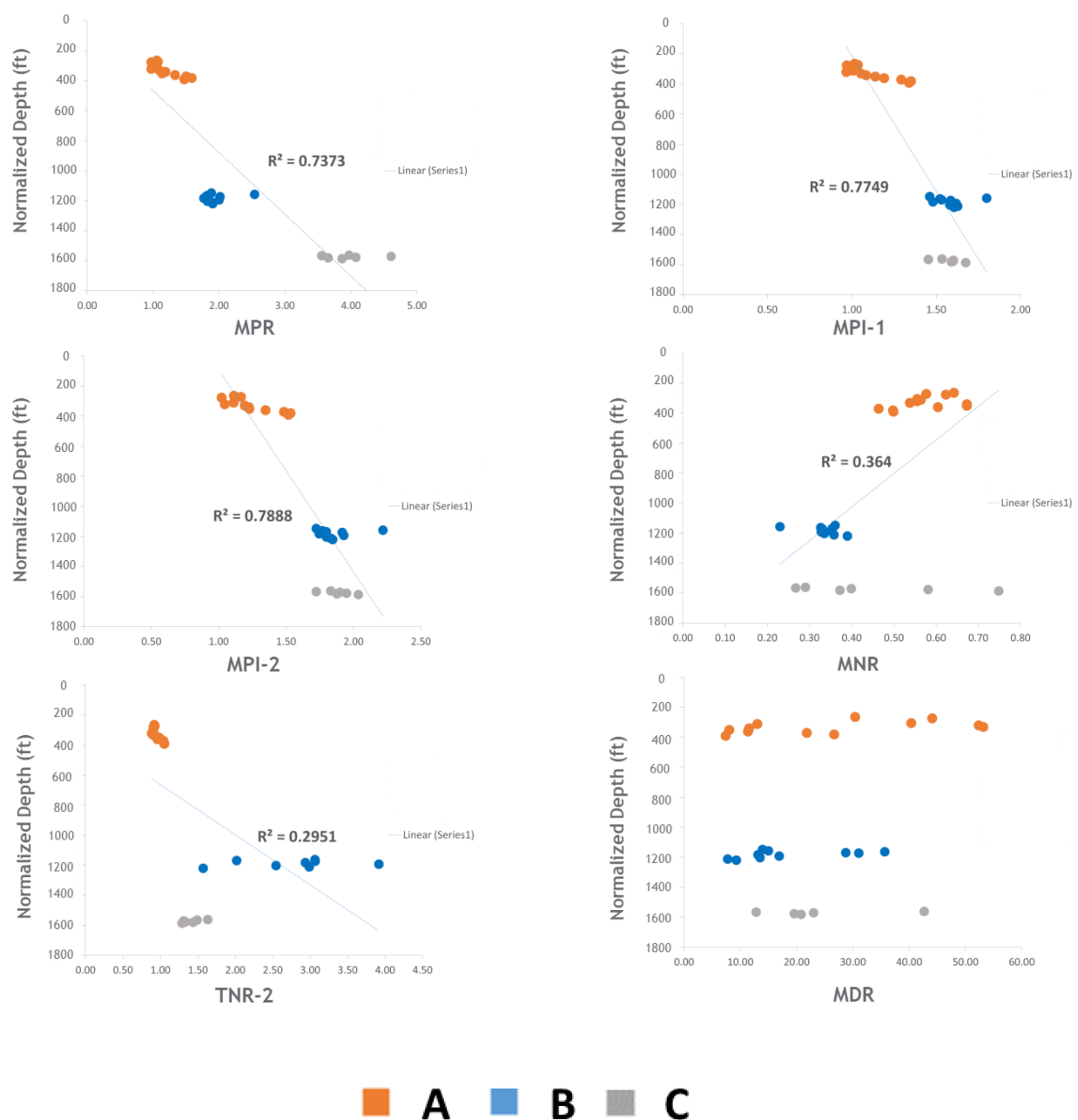


Figure 4—Correlation between selected ratios from each chemical class and depth.

When assessing the correlation with Tmax, the phenanthrene-based ratios again performed well, with MPR showing the best relationship with Tmax, achieving an R^2 of 0.82. In contrast, MNR and TNR-2 exhibited weak correlations with Tmax, with R^2 values below 0.1, while MDR showed no meaningful correlation with Tmax, as illustrated in Figure 5.

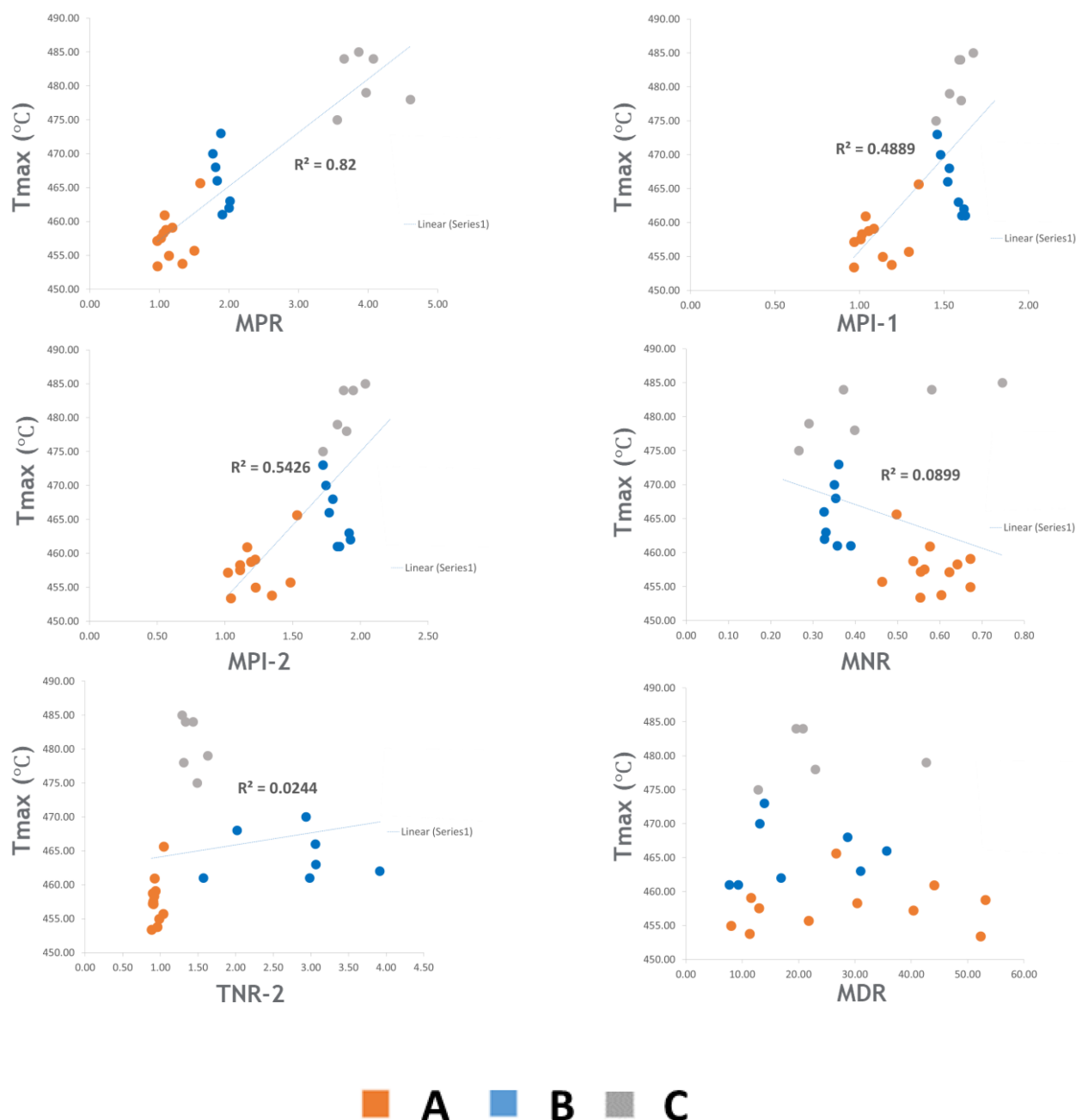


Figure 5—Correlation between selected ratios from each chemical class and T_{max} .

The lack of correlation between MDR, T_{max} and depth highlights the limited reliability of dibenzothiophene-based ratios as maturity indicators in this formation. The distorted values obtained from MDR in samples within the same well can be attributed to the influence of the depositional conditions on the distribution of methyl dibenzothiophenes (Huges 1984) and maturity (Radke et al. 1986). More importantly, the Lower Silurian-age Formation is known for its low abundance of organosulfur within the kerogen, which contributed to challenges in peaks detection and increased uncertainty in the obtained peaks areas.

Based on the results of the previous analyses, multiple published equations were applied to each ratio to calculate the vitrinite reflectance equivalent (%VRE) values. For quality control, the calculated maturity values (%VRE) were plotted against one another, as shown in Figure 6. Although most parameters correctly identified the general maturity window of the samples (Late-Mature), significant variability was observed in the values. The Naphthalene-based ratio MNR generated low maturity values across all wells, indicating it had reached its thermal limit and was thus unreliable at this maturity level. TNR-2 showed more realistic

values but still exhibited lower-than-expected maturity levels and a wide spread of values, particularly in well B, where maturity values ranged from 1.34 to 2.24. The dibenzothiophene ratio (MDR) poses a unique challenge, as it requires the use of two different equations to improve accuracy. The first equation, MDR (I), resulted in a wide statistical spread with unrealistic values (1.05–4.39), while the modified equation, MDR (II), provided slightly improved results but still exhibited a large spread of values in well B and C, as illustrated in Figure 6.

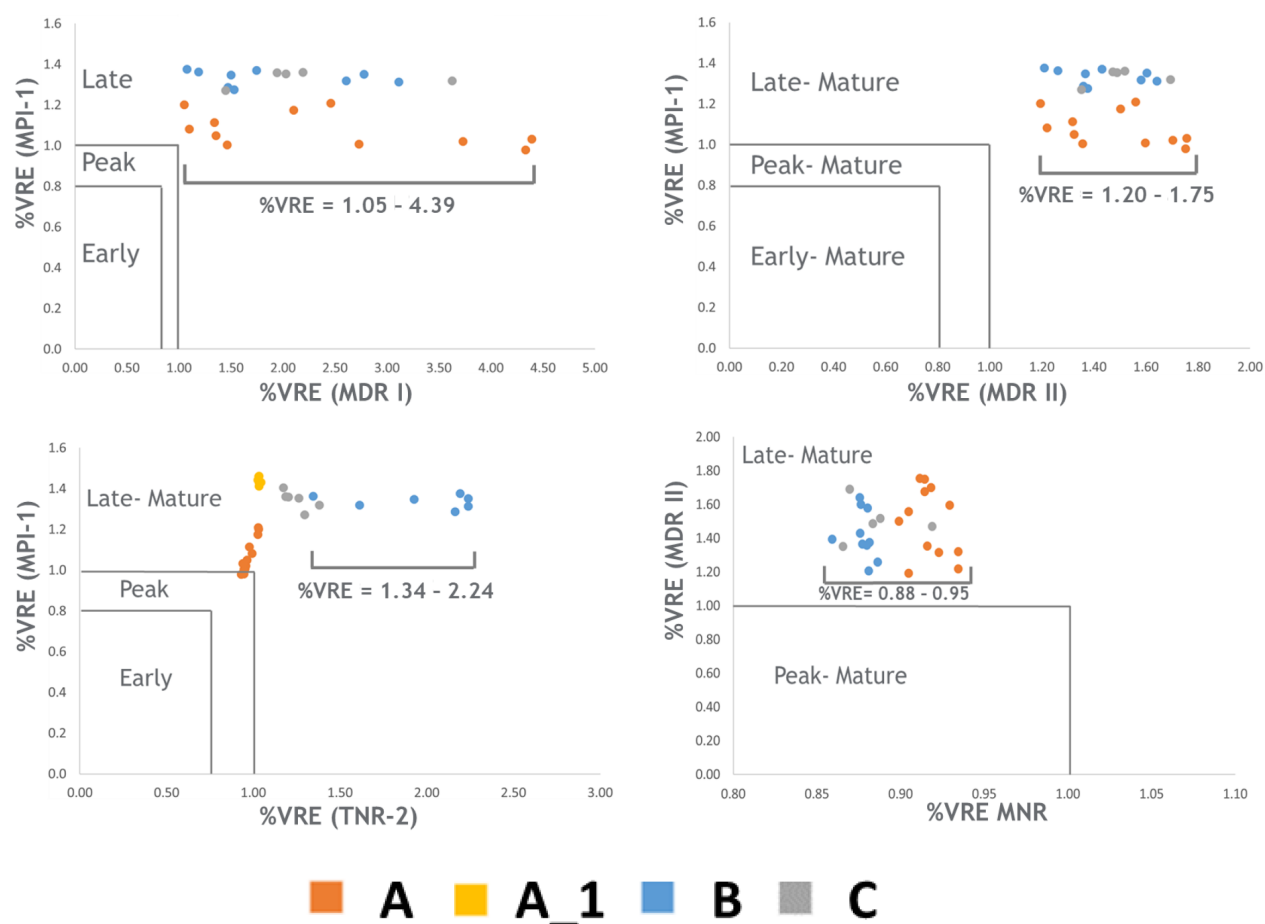


Figure 6—Showing maturity values (%VRE) obtained from selected ratios plotted against each other

From these results, it became evident that phenanthrene-based ratios were the most reliable maturity indicators at this level. These ratios show a consistent distribution of values, reflecting a gradual increase in maturity from well A to C, as shown in Figure 7. Most importantly, the results from phenanthrene-based ratios aligned well with the thermal regime and observed thermal maturity trends in this region (Alraddadi et al., 2024).

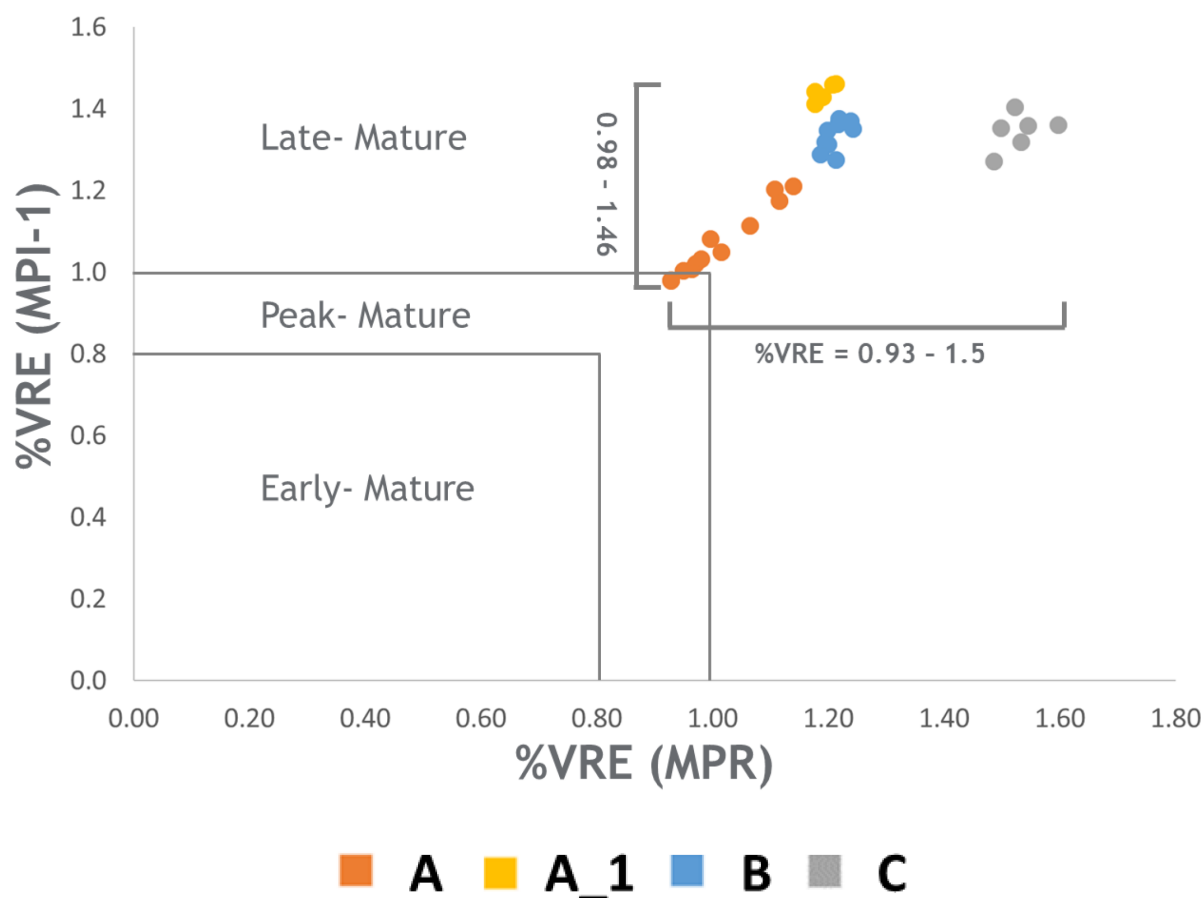


Figure 7—Showing maturity values (%VRE) obtained from two phenanthrenes based ratios, MPI-1 and MPR plotted against each other

Having established that phenanthrene-based maturity estimates are the most reliable, we compared these results with other maturity estimation methods. To do so, we utilized %VRE values obtained from Graptolite, Tmax, and PVT data and compared them with the MPR, MPI-1, and MPI-2 estimates, as illustrated in Figure 8. Overall, graptolite-based values were the highest across all samples, while aromatic-based estimates were generally accurate and aligned with at least one other method. Among the aromatic-based estimates, MPI-2 proved to be the most accurate, whereas MPR appeared to slightly underestimate maturity in some cases.

Finally, maturity estimates based on PVT data, and phenanthrene-based ratios indicated that A_1 fluid samples were more mature than rock samples from the vertical well A. Further investigation is needed to fully understand this discrepancy, but two potential explanations are currently considered. First, rock maturity may be underestimated due to matrix effects, retained fluids or a combination of both, particularly in the graptolite and Tmax-based estimates. Alternatively, the higher maturity of the fluids may result from secondary cracking of retained fluids within the source rock or migration through carrier beds from more mature intervals, as observed in the Permian Basin (Zumberge et al. 2016).

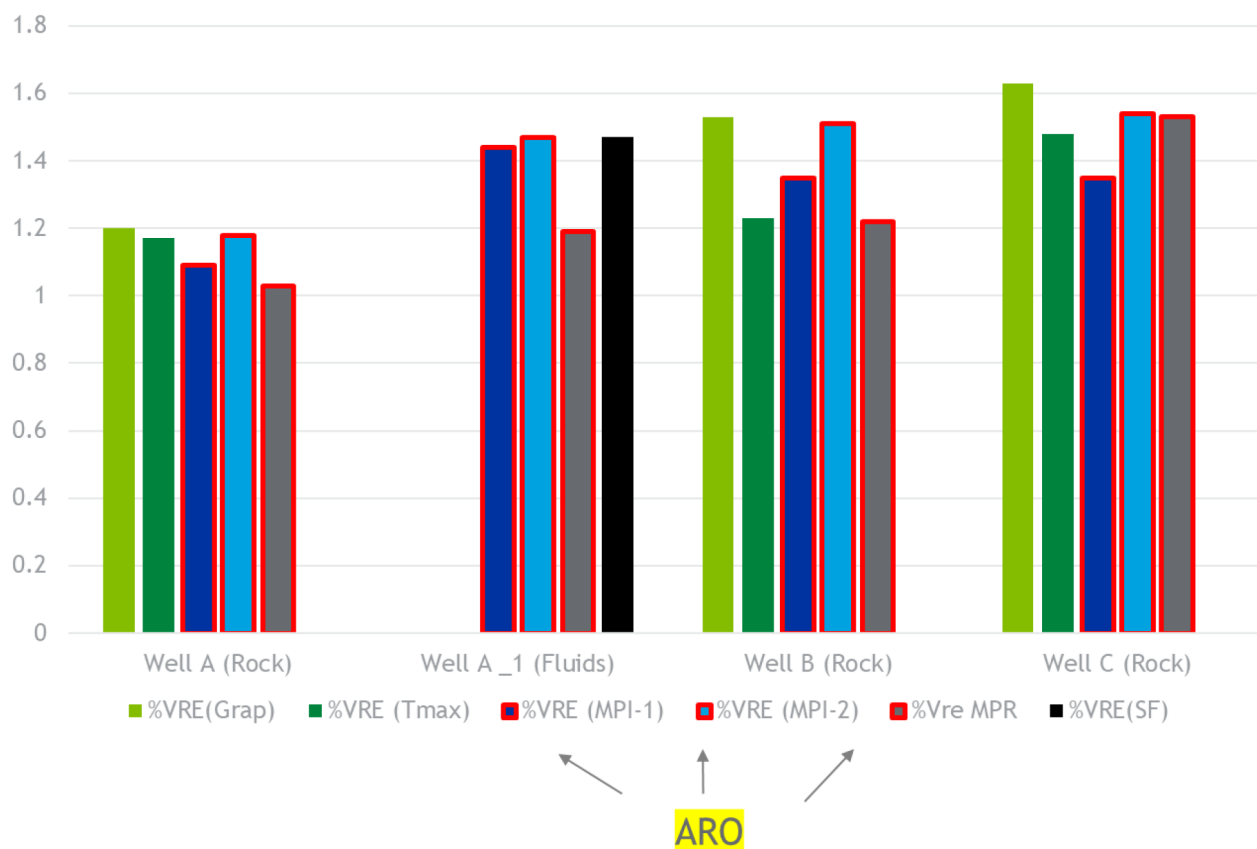


Figure 8—Comparison between averaged maturity values obtained using different methods across all tested wells.

Conclusion

This study highlights both the utility and limitations of different methods for assessing thermal maturity in source rock extracts and generated fluids. Among the tested methods, aromatic hydrocarbons emerged as reliable and versatile indicators of thermal maturity across all tested samples. Utilizing multiple methods can ultimately lead to accurate assessment of thermal maturity in shale source rocks, which is crucial in ensuring the success of unconventional exploration strategies and optimized production planning.

Based on the calculated thermal maturity results, significant variations in values were observed depending on the adopted analytical approach. T-max based maturity estimates underestimate maturity levels in rock samples relative to other tested methods, leading to inaccurate representation of actual maturity levels in this area. This observation aligns with previous studies highlighting the suppression of Tmax due to the influence of fluids retention in this formation. Conversely, graptolite reflectance measurement conducted on rock samples revealed higher maturity values, with a difference of up to 0.3% in comparison to Tmax-derived estimates. For fluid samples, maturity values based on the slope factor of C3-nC5 volatile light hydrocarbons agreed with fluid typing based on gas-to-oil ratio, thus they were utilized as a baseline to assess the variability from other methods. Interestingly, aromatic-based ratios demonstrated remarkable versatility, producing similar maturity values derived from both optical and fluid compositional data analyses, confirming the reliability and consistency of this method in multiple sample types.

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